



Week 4

FOPR111 – Fundamentals of Programming

Identifiers, Variables, Keywords, and Data Types

Objectives:

By the end of this lesson, you should be able to:

- How to write a program in C
- Know the basic structures of C
- Learn the difference between identifiers, variables, keywords, and data types
- Understand variable declaration
- Familiarize with different data types in C



C TOKENS

TOKEN

C programs contain many elements which are identified by the compiler, are known as **tokens**.



For example, consider following program,

```
void main()
{
    int x, y, sum;
    x = 5, y = 10;
    sum = x + y;
    printf ("Sum = %d\n", sum);
}
```

C Tokens can be categorized as follows:

Identifier: main

Special symbols: {}, (,)

Keyword: int

Variables: x, y, sum

Operators: +, =

Tokens: main, {}, (,), int, x, y

Keywords

1. Keywords are pre-defined words (or) reserved words in C compiler.
2. Each keyword is meant to perform a specific function in a C program.
3. Since keywords are referred names for compiler, they cannot be used as identifiers.
4. All keywords must be written in lower case.

In C language there are 32 keywords are available and listed below.

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while

Identifiers

In C language identifiers are the names, which are given to variables, user-defined functions and arrays.

Syntax Rules:

1. An identifier should only have **alphanumeric** characters (**a-z, A-Z, 0-9**) and underscore (**_**) symbol.
2. The first character of an identifier should only have **alphabets (a-z, A-Z)** or underscore (**_**) symbol.
3. Identifiers are case sensitive in C language. For example, “**name**” and “**Name**” are two different identifiers in C language.
4. **Keywords** are not allowed to be used as identifiers.
5. Special characters (Ex: **;**, **,**, white spaces, slash etc.,) are not permitted to use as identifiers.

Comparison between identifier and keywords

Identifier	keyword
Identifiers are names that are given to various program elements, such as variable, function and arrays.	C take some reserved word called keyword, they have predefine meaning in C.
Identifiers consist of letters and digits.	Keyword consist only letter.
Identifier's first character must be a letter.	Keyword's all character is letter.
Identifiers Upper and lowercase letter is use.	Keywords are all lowercase.
Upper and lowercase are not equivalent.	Upper and lowercase are also not equivalent.
Like: X , sum_5 , _weather , 9_name etc. But 4th is not identifier because identifier first character must a letter.	Like: auto, short, long etc.

Variables

A variable is a name that is used to store a value. Variable are simply names used to refer some locations in memory i.e. a location that holds a value.

Variable declaration: Declaration of variable must be done before they are used in the program. It tones three things:

- It tells the compiler what the variable name it is.
- It specifies what type of data the variable will hold.
- It tells the scope or lifetime of a variable.

Variables

Example:

```
int x=10, y=20; // variable initialization
```

```
int sum; // variable declaration
```

```
sum = x+y; // variable definition
```

NOTE: the name of variable must not change, but the value stored in the variable may change during execution of the program.

Rules of variables declaration:

1. All variables must be declared before they can be appeared in executable statements.
2. A variable name must not start with digits.
3. Keywords are not allowed as variable names.
4. White spaces are not allowed in variable names.
5. Variable names can consists of alphabets, digits and only permitted special symbol underscore (_).

Data types in C Language

Type	Format	Meaning	(Example(s))
int	%d or %i	-a whole number	(3, <u>7</u> , <u>+8</u> , <u>-9</u> , 1000000)
float	%f	-a number with decimal point (5.8, 9.0, -5.6789, +0.0032)	
char	%c	-a single letter, symbol, or number enclosed within two single quotes (<code>'B'</code> , <code>'r'</code> , <code>'*'</code> , <code>'3'</code>)	
char	%s	-a string in C is considered as a series of characters (<u>"Ms."</u> , <u>"Hi!"</u> , "143", "I love")	
double	%f	-for bigger or larger numbers with a decimal point (123456789.123)	

Thank you

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